

pretty (and smart)

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outline

looks matter

general: displaying data (in a map)

maps specifically

but wait, smart is important too! see [final_project.pdf](#) [later]

examples from past [if time]

today

- switching gears and relaxing: art of mapping

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“looks vs brains”

- of course the key, say 50% of success is:
great idea, 'the story'
- and also very important, say another 40%:
 - good/quality data: GIGO!
 - the right vars (eg measurement, standardization)
 - at the right level (resolution or aggregation)
- but if map's ugly:
people won't bother to get its “internal beauty”
- mapping is about the “visual appearance”
it has to be pretty! “the looks” come first!

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make it interesting!

- “the greatest value of a picture is when it forces us to notice what we never expected to see”
- if a dog bites a man it's no news,
if a man bites dog, that's news
- or less poetically, the maps need to make the point,
tell a story, ideally unexpected, fascinating, etc
 - and in a convincing (and rigorous) way
- ref: Tufte (multiple)
<http://www.edwardtufte.com/tufte/>

be simple

- make everything as simple as possible, but not simpler
- no padding: present only data needed for the story
- no clutter: put in a single fig/map only the data that are highly related and must be compared
- not super relevant but may be useful? put in app/som/ipynb
 - people looking for extra info will find it
 - people interested in the main story won't get distracted

avoid visual clutter

- all parts of fig must be meaningful
- eg dont use:
 - shades
 - fancy colors
 - any decoration

use space efficiently

- get rid of white space
- blow up your map as big as possible, use all space
- put legend in an efficient way
- eg in water or forest, etc

balance! keep balance in:

- colors (say either use toned down colors for everything or use stark contrasts, etc)
btw colors have meaning! eg red v green; yellow is happy, green calming <https://towardsdatascience.com/the-crucial-role-of-color-theory-in-data-analysis-and-visualization/>
- fonts: titles, notes, labels etc should be proportional
- thickness of lines
- and everything else
- in general:
 - rather use less ink than more
 - spend ink on what matters, the story

one v several graphs/maps

- to convey an idea (tell a story) usually need several maps
- say to show deprivation in NJ:
 - low educ, poverty, crime, etc
 - can also map change: better just 1 map than 2
just calc chng var: $100 * (\text{pop10} - \text{pop00}) / \text{pop00}$

meaning! think about it! the 'so what?' question

- ok, you've got the map...now think about it
 - (beyond technical correctness; lack of mistakes)
- what does it say?
 - is it interesting, informative, or helpful?
 - as you look at it, ask yourself the 'so what?' question
 - if not happy with the answer:
 - move it to the app/som, and produce a better map
- i will be grading substantive meaning, too

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some examples [if time]

- we can have a look at some examples and discuss if they are pretty (and smart/interesting)
- <http://twistedsifter.com/2013/08/maps-that-will-help-you-make-sense-of-the-world/>
- also do “map lit rev” ie google images “what you study” + “map” eg “healthy food stores philly map,” “industry clusters map”

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nj force [only on campus or vpn!]

- started in this class:

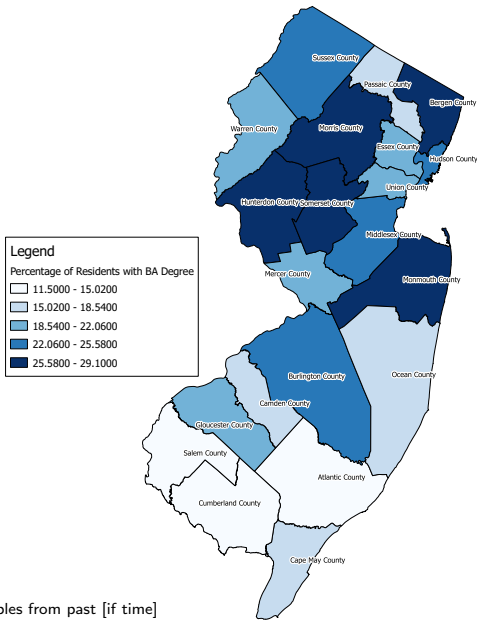
`https://maps-dppa.camden.rutgers.edu/index.php?search=ethan&title=Special%3ASearch&go=Go`

- and we published a paper:

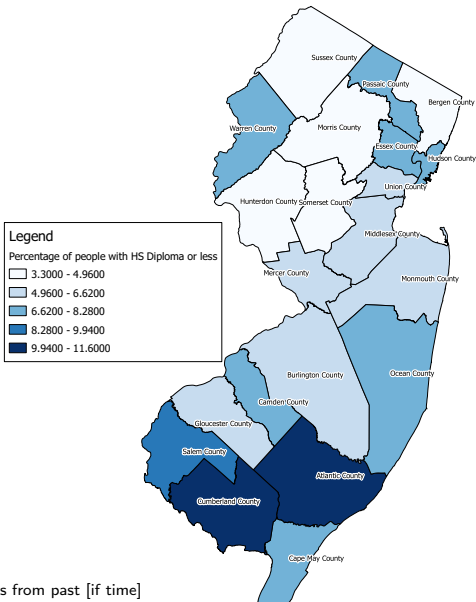
`https://www.sciencedirect.com/science/article/pii/S0047235221000489`

- next story telling about education in NJ with several maps

New Jersey Residents with Bachelor Degree or Higher



New Jersey rate of residents earning less than a High School Diploma



2012-2013 NJ K-12 Education Aid per resident

Legend

NJ State Aid Per Resident 2012-2013

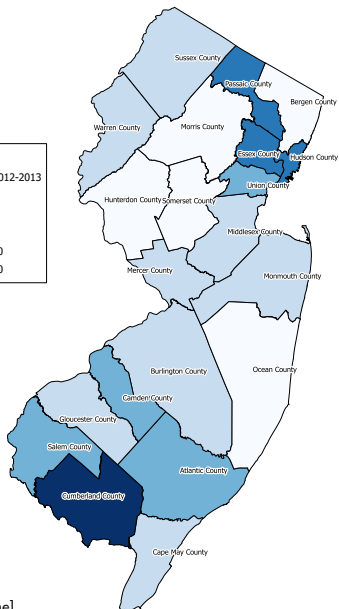
226.0000 - 609.6000

609.6000 - 993.2000

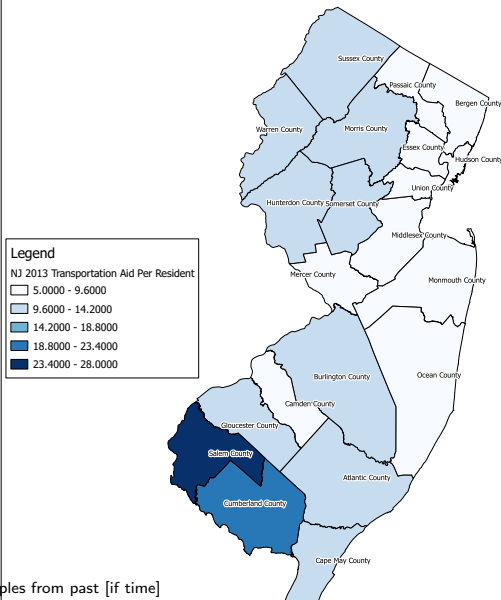
993.2000 - 1376.8000

1376.8000 - 1760.4000

1760.4000 - 2144.0000



NJ 2013 Transportation Aid Per Resident

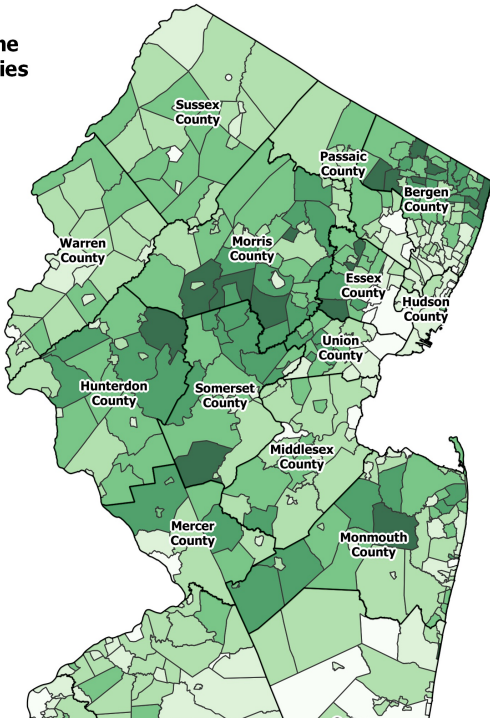
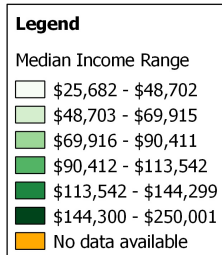


NJ counties

- many variables about similar topic: education
- great: triangulation
- good use of space, could be little better
- nice color ramp
- good fonts, maybe title little smaller
- fewer decimal points !
- could list data source (but may do it elsewhere, say in paper)

Median Household Income in New Jersey Municipalities

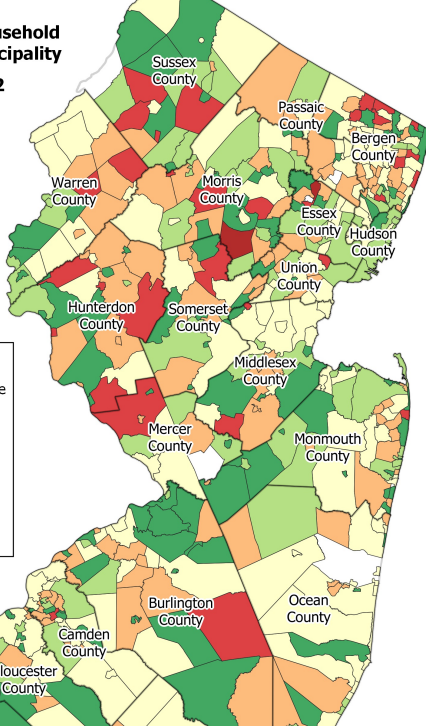
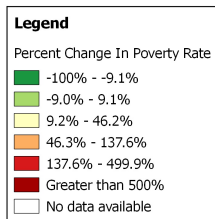
For 2006-2010
Median Income: \$67,681



income in NJ

- note, map cut to display detail
- nice title, we know timeframe
- legend: note decimal points; love yellow for missing data
- really nothing to fix; 2 things below *personal* preference:
 - use white borders for municipalities—little cleaner
 - use a 2 color ramp (say red-green) to signal poverty-affluence
- interesting: such big disparities so close to each other
- a note on level of analysis:
 - if county level data: N dark green; S: bright green
 - it would cover up deep poverty pockets in NYC metro!
- aggregation does matter! results are totally different

**Percent Change in Household
Poverty Rates by Municipality**
From 1999-2012



poverty change

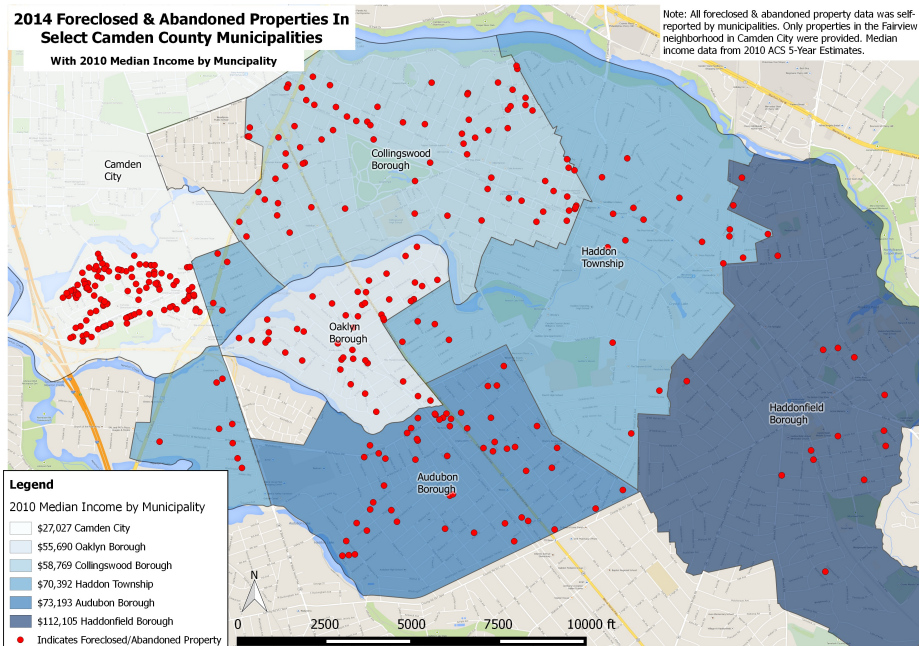
- readable, informative, great!
- great variation ! quite striking
- wonder why some places do quite well and some so bad...
- future: change from 1% to 5% and from 10% to 50% are both 500%
- yet they mean different things
- can use it for policy analysis...
- say could thing why such great divergence...
- any policies that could have worked?

properties

2014 Foreclosed & Abandoned Properties In Select Camden County Municipalities

With 2010 Median Income by Municipality

Note: All foreclosed & abandoned property data was self-reported by municipalities. Only properties in the Fairview neighborhood in Camden City were provided. Median income data from 2010 ACS 5-Year Estimates.



Legend

2010 Median Income by Municipality

- \$27,027 Camden City
- \$55,690 Oaklyn Borough
- \$58,769 Collingswood Borough
- \$70,392 Haddon Township
- \$73,193 Audubon Borough
- \$112,105 Haddonfield Borough

• Indicates Foreclosed/Abandoned Property

examples from past [if time]

properties

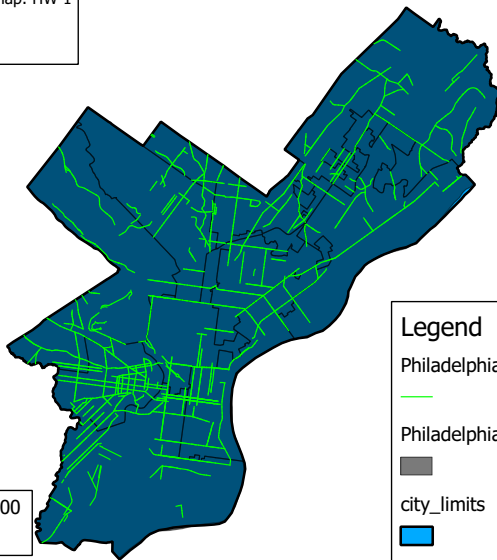
- superb use of space! labeling neatly in corners
- pretty map!
- interesting map—clearly foreclosures correlate negatively with income

properties

- guess some data missing in top right corner? should be indicated more clearly...
- perhaps make it reddish or cover with dashed or dotted pattern
- so that we know clearly for which areas there are no data
- 2014 is still going—should mention months
- note how hi resolution google map is! can zoom in and see all road names
- is it median hh income? (it is explained in journal though)

bike lanes

Philadelphia bike map: HW 1



Legend

PhiladelphiaBikeNetwork201204



PhiladelphiaCouncilDistricts_2000



city_limits



0 20000 40000



bike lanes

- very nice !
- can improve: beautify fonts, colors...
- future research:
 - add bike traffic (not sure if available, but can measure...)
 - add sociodemographics by tract
 - eg do young or rich or educated people bike ?
 - compare with other cities, eg Boston, Portland...

future ps

- we'll be doing thematic maps throughout the course
- and i will be much more picky